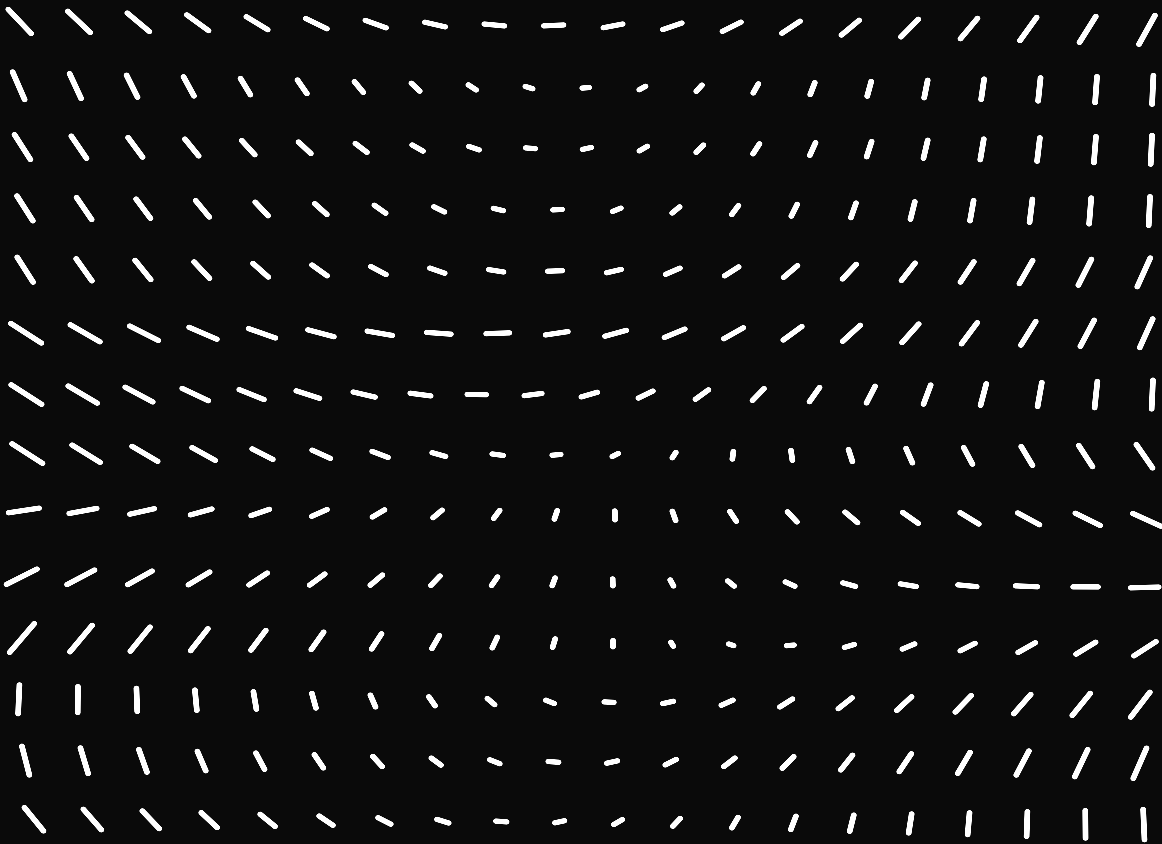


Strogatz thinks thus the systems, in general, follows:

Number of variables $\longrightarrow$

	$n = 1$	$n = 2$
linearity	Growth / Decay Exponential growth RC Circuit Radiactive decay	Oscillations linear Osc Mass & Spring RLC Circuit 2-body Problem.
Non-linearity	· Fixed points · Bifurcations · Overdump Systems, Relaxational Dynam. · Logistic eq. for popn	Pendulum Anharmonic oscillators Limit cycles Biological oscillators (neurons - heart cells) Predador - Prey modells Non-linear electronices (von der Pol, Josephson)

So, at first he asks to start with the most simple possible system: $n = 1$. Then we get the eqn. $x' = f(x)$. We call this kind of systems **first order systems** or one dimensonal flows.

To be clear;

1.- Here "system" is used in the sense of dynamical system instead of the one in classical mechanics, here one single eqn could be a system.

2.- For symplicitly thru all this chapter, unless explicit other way we will **always think in autonomus systems**. I.e. $f(x) : \neq f(x,t)$.

2.1 A Geometric Way of thinking

One of the most powerful ideas in dynamical systems (introduced by my personal hero: Poincaré) is the one of thinking the differential equation as a vector field. This idea is that powerful because help us to understand the long-term behavior of the systems without the need of a closed solution/analytical solution.

Let's consider the following system: $\dot{x} = \sin(x) \quad \ldots (1)$
Such equation could be solved by means of integration

$$\dot{x} = \sin x \Rightarrow \frac{1}{\sin x}\frac{dx}{dt} = 1 \Rightarrow \int \frac{1}{\sin x}\frac{dx}{dt}dt = \int dt$$

$$\Rightarrow \int \frac{1}{\sin x}\frac{dx}{dt} = t + cte, \quad \text{si} \quad \hat{x} = x \quad \text{then} \quad d\hat{x} = \frac{dx}{dt}dt$$

$$\Rightarrow \int \frac{1}{\sin(\hat{x})}d\hat{x} = t + cte, \quad \text{If} \quad I(x) = \int \frac{1}{\sin x}dx \quad \text{then we have}$$

$$\frac{1}{\sin x} = \csc(x)$$

$$I(x) = \int \frac{1}{\sin x}dx = \int \csc(x)\,dx \quad \text{if we multiply by} \quad \frac{\cot(x) + \csc(x)}{\cot(x) + \csc(x)}$$

$$= \int \frac{\csc(x)(\cot x + \csc x)}{(\cot x + \csc x)}dx = \int \frac{\csc^2 x + \csc x \cot x}{\cot x + \csc x}dx$$

. now let use (-1)

$$= \int - \frac{-\csc^2 x - \csc x \cot x}{\cot x + \csc x}dx$$

if we make $u = \cot x + \csc(x)$
then $du = (-\csc^2 x - \cot x \csc x)\,dx$

$$= \int - \frac{du}{u} = -\ln(u) + cte = -\ln(\cot x + \csc(x)) + cte_f$$

Now, we could use this result in the last eqn.

We have:

$$\int \frac{1}{\sin \hat{x}}\, d\hat{x} = t + cte \text{ , using the result for } I(x)$$

$$-\ln\left(\cot(x) + \csc(x)\right) = t + cte$$, multiplying by (-1) and exp both sides

$$\cot(x) + \csc(x) = e^{-t+cte} = x_0 e^{-t}$$

$\therefore$ $\cot x + \csc x = x_0 e^{-t}$ is the solution to $x' = \sin x$

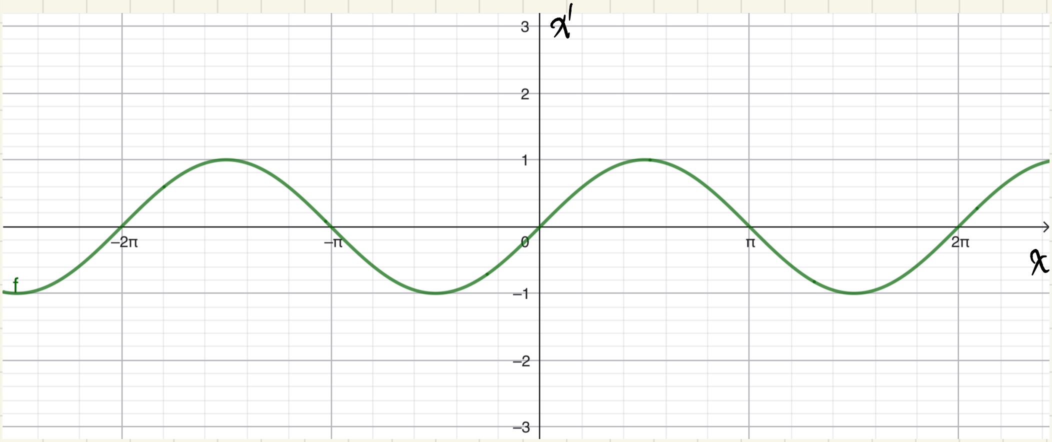
Well the big deal is that the last equation is non-usable, we cannot easily to answer:

1. if $x_0 = \pi/4$, what happens when $t \to \infty$?

2. for any arbitrary condition x_0, what is the behavior of $x(t)$ as $t \to \infty$?

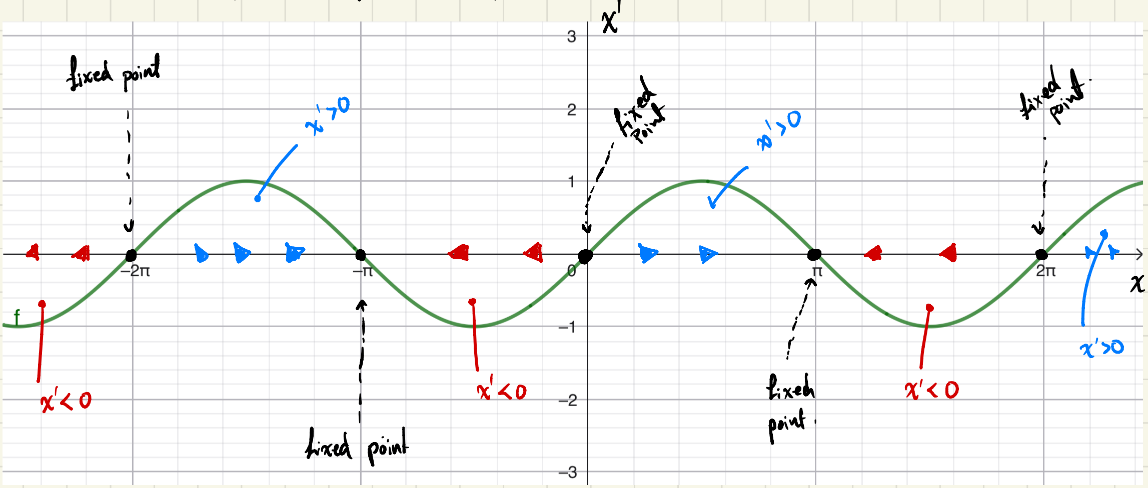
Well... It's not transparent, for not saying, not useful.

Now, let's think in the differential equation as a vector field and a geometric entity:



Let's notice that $f(x)$ is the relation between x & x'

Now if, for example, $x_0>0$ but $x_0<\pi$ the $x'>0$.
On the other hand, if $x_0<0$ but $x_0>-\pi$ then $x'<0$
What means this? Well, if $x'>0$ for x_0 then $x(t)$ will grow.
On the other hand, if $x'<0$ for x_0 then $x(t)$ will contract.
But what if $x_0 = n\pi$ (for $n\in\mathbb{Z}$), well $x'=0$ and then $x(t)$ will be a steady-state / stationary on x_0.



This last picture gift us a superpower, the one of see the future, for example:

- if $x_0\in(0,\pi)$ then $x(t)\to\pi$ as $t\to\infty$
- if $x_0\in(-\pi,0)$ then $x(t)\to-\pi$ as $t\to\infty$
- if $x_0=n\pi$ $(n\in\mathbb{Z})$ then $x(t)=x_0$ $\forall t$

This power is absolute & 100% certainty. If we know, we know.

So, now we can answer the questions:

1. If $x_0=\pi/4$ then $x(t)\to\pi$ as $t\to\infty$

2. If $n \in \mathbb{Z}^+$ then if $x \in (n\pi, (n+1)\pi)$ and n even
$x(t) \longrightarrow (n+1)\pi$ as $t \to \infty$. If n is odd then
$x(t) \longrightarrow n\pi$ as $t \to \infty$

On the other hand, if $n \in \mathbb{Z}^- \cup \{0\}$ then
if $x_0 \in ((n-1)\pi, n\pi)$ an n is even then
$x(t) \longrightarrow (n-1)\pi$ as $t \to \infty$, if n is odd then
$x(t) \longrightarrow n\pi$ as $t \to \infty$.

Everything without using the exact (implicit) solution.

2.2 Fixed Points & Stability

So, for make my life